

# Midterm

## Solutions

# Problem 1

Let  $C_i = \begin{cases} 1 & \text{if red} \\ 0 & \text{if white} \end{cases}$  for  $i = A, R, Y. \Rightarrow$  a state  $\omega = C_A C_R C_Y$  is a sequence of 0, 1's

1.a  $\Omega = \{001, 010, 100, 110, 101, 011, 111\}$

1.b  $P_Y = \{\{001\}, \{010, 011\}, \{100, 101\}, \{110, 111\}\}$

$P_R = \{\{001, 010, 011\}, \{100, 110, 101, 111\}\}$

$P_A = \{\{001, 010, 100, 110, 101, 011, 111\}\}$

1.c.  $P_R = \{\{001\}, \{010, 011\}, \{100, 110, 101, 111\}\}$

$P_A = \{\{001\}, \{010, 100, 110, 101, 011, 111\}\}$

Verbal saying "No" indicates the state cannot be  $\omega \notin \{001\}$ , i.e.  $\omega \in \{001\}^c$

1.d.  $P_A = \{\{001\}, \{010, 011\}, \{100, 110, 101, 111\}\}$

At any  $\omega \in \{010, 011\}$  Rysky would have been able to say "Yes, I know". Thus, her answer indicates that  $\omega \in \{010, 011\}^c$

In each of these partition elements, Annur's hat color is the same  $\Rightarrow$  she is able to tell her color always!

the actual hat configuration.

1.e. Given the "No", "No" answers, Annur is able to say that her hat is **RED**. The answer depends on the true state.

Since we did this **without specifying the true state of the world**, the fact that she can answer is always true, no matter what is the actual hat configuration. As can be seen from  $P_A$  in 1.d. Annur is always able to answer.

Problem 2 Since I was not asking anything specific about the sequence of opinions (time to convergence, what they say at each stage, etc.), it is sufficient to appeal to Thm 1.30 from my slides (Georghiopoulos & Polemarchakis 82) to conclude that they will indeed say the same thing at the end of the discussion.



When you really know how to apply the results confidently, you can be really FAST.

If you decided to do it the long way (which in this case is not very long), here it is.  
 (Note, there is a different approach - defining  $A_k = \{\omega \in \Omega : x(\omega) = k\}$  for  $k = 1, 0, -7$  and computing potentials

$$Q_1^0(\omega) = \begin{cases} 1 & \omega = 1, 2, 3 \\ 10/3 & \omega = 4, 5, 6 \\ -14/3 & \omega = 7, 8, 9 \end{cases}$$

$$Q_2^0(\omega) = \begin{cases} 3/4 & \omega = 1, 2, 3, 4 \\ -1 & \omega = 5, 6, 7, 8 \\ 0 & \omega = 9 \end{cases}$$

$$Q_1^1(\omega) = \begin{cases} 1 & \omega = 1, 2, 3 \\ 0 & \omega = 4 \\ 5 & \omega = 5, 6 \\ -7 & \omega = 7, 8 \\ 0 & \omega = 9 \end{cases}$$

They are  
the same.

$$Q_2^1(\omega) = \begin{cases} 1 & \omega = 1, 2, 3 \\ 0 & \omega = 4 \\ 5 & \omega = 5, 6 \\ -7 & \omega = 7, 8 \\ 0 & \omega = 9 \end{cases}$$

### Problem 3

3.a. The players are symmetric so what we conclude for one also works for the other.

$$\begin{aligned} I \text{ is dominant in } \Theta_G &\Leftrightarrow z+1+y_G > z \text{ and } 1 > 0 \\ &\Leftrightarrow 1+y_G > 0 \Leftrightarrow y_G > -1 \end{aligned}$$

$$\begin{aligned} N \text{ is dominant in } \Theta_B &\Leftrightarrow z > z-1+y_B \text{ and } 0 > -1 \\ &\Leftrightarrow 0 > -1+y_B \Leftrightarrow 1 > y_B \end{aligned}$$

3.b. Assume  $z > 1$   $\left\{ \begin{array}{l} \text{When } \Theta_B : 2(z-1+y_B) > z-1 \Leftrightarrow z-1+2y_B > 0 \Leftrightarrow z > 1-2y_B \\ \text{When } \Theta_G : 2(z+1+y_G) > z+1 \Leftrightarrow z+1+2y_G > 0 \Leftrightarrow z > -1-2y_G \end{array} \right.$

3.c.  $\Omega = \mathbb{H} \times T_1 \times T_2$  will have 8 states but some of them will have probability zero, so we can simplify this and not consider them. With this observation, the belief space is:

$$\Omega = \{(\Theta_B, t_b, t_b), (\Theta_G, t_g, t_g), (\Theta_B, t_g, t_g)\} \quad T_i = \{t_g, t_b\}, \quad \tau_i: \Omega \rightarrow T_i$$

where  $\tau_i(\Theta, t_1, t_2) = t_i, \quad i=1,2;$

$$\mu(\Theta, t_1, t_2) = Pr(\Theta) \cdot \pi(t_1, t_2 | \Theta) \rightarrow \begin{cases} \mu(\Theta_B, t_b, t_b) = (1-\psi)q \\ \mu(\Theta_B, t_g, t_g) = (1-\psi)(1-q) \\ \mu(\Theta_G, t_g, t_g) = \psi \end{cases}$$

Let's first compute the posteriors: Using Bayes Rule

$$\mu_i(\Theta_B | t_b) = \frac{\mu(\Theta_B, t_b, t_b)}{\mu(\Theta_B, t_b, t_b)} = 1$$

$$\mu_i(\Theta_G | t_b) = 0$$

$$\mu_i(\Theta_G | t_g) = \frac{\mu(\Theta_G, t_g, t_g)}{\mu(\Theta_G, t_g, t_g) + \mu(\Theta_B, t_g, t_g)} = \frac{\psi}{\psi + (1-\psi)(1-q)}$$

$$\mu_i(\Theta_B | t_g) = \frac{\mu(\Theta_B, t_g, t_g)}{\mu(\Theta_G, t_g, t_g) + \mu(\Theta_B, t_g, t_g)} = \frac{(1-\psi)(1-q)}{\psi + (1-\psi)(1-q)}$$

Note: A player that receives signal  $t_b$  knows that the state is  $\Theta_B \Rightarrow$  has a dominant strategy to play N.

On the other hand, players that receive signal  $t_g$ , deem possible  $\Theta_B$  or  $\Theta_G$ .

We will find out below how large  $q$  (interpreted as the accuracy of the signal) needs to be to make I under  $t_g$  a BNE.

3.d.  $S_i(t_g) = I$ ,  $i=1,2$  is an equilibrium if the following is true:

There is no profitable deviation from  $I$  given that the other player is investing

$$\Leftrightarrow \underbrace{\frac{\psi}{\psi + (1-\psi)(1-q)} [z + 1 + y_A]}_{\text{Payoff under } \Theta_A \text{ when both } I} + \underbrace{\frac{(1-\psi)(1-q)}{\psi + (1-\psi)(1-q)} [z - 1 + y_B]}_{\text{Payoff under } \Theta_B \text{ when both } I} > \underbrace{\frac{\psi}{\psi + (1-\psi)(1-q)} z}_{\text{Payoff deviating under } \Theta_A} + \underbrace{\frac{(1-\psi)(1-q)}{\psi + (1-\psi)(1-q)} z}_{\text{Payoff deviating under } \Theta_B}$$

since  $\psi + (1-\psi)(1-q) > 0$  the above is equivalent to:

$$\psi(z + 1 + y_A) + (1-\psi)(1-q)(z - 1 + y_B) > \psi z + (1-\psi)(1-q)z$$

$$\Leftrightarrow \cancel{\psi z + (1-\psi)(1-q)z} + \psi(1 + y_A) + (1-\psi)(1-q)(-1 + y_B) > \cancel{\psi z + (1-\psi)(1-q)z}$$

$$\Leftrightarrow \psi(1 + y_A) + (1-\psi)(1-q)(-1 + y_B) > 0$$

$$\Leftrightarrow q(1 - y_B)(1-\psi) > (1-\psi)(1 - y_B) - \psi(1 + y_A), \quad \text{using } 0 < \psi < 1 \text{ and } y_B < 1, y_A > -1$$

$$\Leftrightarrow q > 1 - \frac{\psi}{1-\psi} \frac{1 + y_A}{1 - y_B}$$

3.e. The condition is exactly the same as above, again, assuming